

Testing the alg Lcs problem

$X = \langle A, B, C, D \rangle$ $m=4$
 $Y = \langle A, C, D \rangle$ $n=3$
 $\rightarrow j$

		0	1	2	3
		A	C	D	
0	0	0	0	0	0
1	A	0	1	1	1
2	B	0	1	1	1
3	C	0	1	2	2
4	D	0	1	2	3

$i = 1$ to 4 $i=1$

$j = 1$ to 3 $j=1$

if $(x_i == y_j)$

A A

$c[1,1] = c[0,0] + 1$

$b[1,1] = "\uparrow"$

$j=2$

if $(x_i \neq y_j)$

A C

if $c[0,2] < c[1,1]$

0

$c[1,2] = c[1,1]$

$b[1,2] = "\leftarrow"$

$j=3$

if $(x_i \neq y_j)$

A D

if $c[0,3] < c[1,2]$

0 < 1

$c[1,3] = c[1,2]$

$b[1,3] = "\leftarrow"$

$i=2$

$j = 1$ to 3 $j=1$

if $x_i \neq y_j$

B A

if $c[1,0] \geq c[2,0]$

1 > 0

$c[2,1] = c[1,1]$

$b[2,1] = "\uparrow"$

(Dynamic programming)

$j=2$

if $x_i \neq y_j$

B C

if $c[1,2] \geq c[2,1]$

$c[2,2] = c[1,2]$

$j=3$

if $x_i \neq y_j$

B D

if $c[1,3] \geq c[2,2]$

1 1

$c[2,3] = 1$

$b[2,3] = "\uparrow"$

$i=3$

$j = 1$ to 3 $j=1$

if $x_i \neq y_j$

C A

if $c[2,1] \geq c[3,0]$

1 > 0

$c[3,1] = c[2,1]$

$b[3,1] = "\uparrow"$

$j=2$

if $(x_i == y_j)$

C C

$c[3,2] = c[2,1] + 1$

$b[3,2] = "K"$

$j=3$

if $x_i \neq y_j$

C D

if $c[2,3] < c[3,2]$

1 < 2

$c[3,3] = 2$

$b[3,3] = "\leftarrow"$

$i=4$

$j = 1$ to 3 $j=1$

if $x_i \neq y_j$

D A

~~$c[4,1] = c$~~

if $c[3,1] > c[4,0]$

1 > 0

$c[4,1] = 1$

$b[4,1] = "\uparrow"$

$j=2$

if $x_i \neq y_j$

D A

$c[4,2] = ?$

if $c[3,2]$

$c[4,2] = ?$

$b[4,2] = ?$

$j=3$

if $x_i \neq y_j$

D A

$c[4,3] = ?$

$b[4,3] = ?$

$j=2$
 if $x_i \neq y_j$
 $D \neq C$
 if $C[3,2] > C[4,1]$
 $C[4,2] = C[3,2]$
 $b[4,2] = \uparrow$
 $j=3$
 if $x_i \neq y_j$
 $D \neq D$
 if $C[4,3] = C[3,2] + 1$
 $= 3$
 $b[4,3] = \nwarrow$

Soln

	A	C	D
0	0	1	2
1	0		
2	1		
3	2		
4	3		

 $C[4,3] = 3$ (length of the common subsequence)
 If you print the characters corresponding to " $\nwarrow$ " (diagonal) arrow, we will get the common subsequence
 $b[4,3] = \nwarrow \Rightarrow x_4 = y_3 = A$
 $b[3,2] = \nwarrow \Rightarrow x_3 = y_2 = C$
 $b[1,1] = \nwarrow \Rightarrow x_1 = y_1 = A$
 So common subsequence = $\langle A, C, A \rangle$

Matrix chain multiplication using dynamic programming

Tracing the alg

Let there be four matrices

A , B , C and D
 (5×3) (3×2) (2×3) (3×4)

Let there be 2 tables

$p_0 = 5$ (no of rows in A)
 $p_1 = 3$ (no of cols in A)
 $p_2 = 2$ (no of cols in B)
 $p_3 = 3$ (no of cols in C)
 $p_4 = 4$ (no of cols in D)

		$m \rightarrow j$						j			
			1	2	3	4			2	3	4
i	1	0	30	60	94		i	1	1	2	2
	2		0	18	48			2		2	2
	3			0	24			3			3
	4				0						

$n=4$

$l \leftarrow 2$ to 4 $l=2$

$i \leftarrow 1$ to 3 $i=1$

$j=2$

$k=1$

$m(1,2) = 2$

$$m(1,2) = 30$$

$$s(1,2) = k=1$$

$$i=2$$

$$j=3$$

$$k=2$$

$$q = m(2,2) + m(2,3) + p_1 p_2 p_3 = 0 + 0 + 3 \times 2 \times 3 = 18$$

$$m(2,3) = 18, s(2,3) = 2$$

$$i=3$$

$$j=4$$

$$k=3$$

$$q = m(3,3) + m(4,4) + p_2 p_3 p_4 = 0 + 0 + 2 \times 3 \times 4 = 24$$

$$m(3,4) = 24, s(3,4) = 3$$

$$l=3$$

$$i=1 \text{ to } 2 \quad i=1$$

$$j=3$$

$$k=1, 2$$

$$q = \begin{cases} k=1, m(1,1) + m(2,3) + p_0 p_1 p_3 = 0 + 18 + 5 \times 3 \times 3 = 63 \\ k=2, m(1,2) + m(3,3) + p_0 p_2 p_3 = 30 + 0 + 5 \times 2 \times 3 = 60 \end{cases}$$

$$m[1,3] = 60, s[1,3] = 2$$

$$i=2$$

$$j=4$$

$$k=2 \text{ to } 3$$

$$q = \begin{cases} k=2, m(2,2) + m(3,4) + p_1 p_2 p_4 = 0 + 24 + 3 \times 2 \times 4 = 48 \\ k=3, m(2,3) + m(4,4) + p_1 p_3 p_4 = 18 + 0 + 3 \times 3 \times 4 = 54 \end{cases}$$

$$m[2,4] = 48, s[2,4] = 2$$

$$l=4$$

$$i=1 \text{ to } 1 \quad i=1$$

$$j=4$$

$$k=1 \text{ to } 3$$

$$q = \begin{cases} k=1, m(1,1) + m(2,4) + p_0 p_1 p_4 = 0 + 48 + 5 \times 3 \times 4 = 108 \\ k=2, m(1,2) + m(3,4) + p_0 p_2 p_4 = 30 + 24 + 5 \times 2 \times 4 = 94 \\ k=3, m(1,3) + m(4,4) + p_0 p_3 p_4 = 60 + 0 + 5 \times 3 \times 4 = 120 \end{cases}$$

Soln is $m(1,4) = 94$ [min no of scalar multiplications needed to compute product of 4 matrices]

$s(1,4) = 2 \Rightarrow$ The paranthesis should be after 2nd matrix.

So $((AB)(CD))$ is the soln.